Preprocessing & Feature Engineering Report: Shifting Tornado Hazard in the United States and Emerging Insurance Exposure

Merrimack College
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1 Recap Background and Questions

This project examines whether tornado hazard in the United States is shifting from the traditional Great Plains Tornado Alley toward the Southeast's Dixie Alley region over time, and whether counties predicted to exhibit elevated tornado hazard in the following year also have greater present-day insurance exposure based on population, housing units and median home value.

The hypothesis is that tornado hazard is shifting from the Tornado Alley region toward Dixie Alley due to increasing tornado frequency and severity in the Southeast over time. It is predicted that counties located in Dixie Alley will be more likely classified as high tornado hazard in the following year compared to Tornado Alley and will exhibit a higher present-day insurance exposure.

2 Methods

2.1 Previous Plan and Adjustments

The initial preprocessing plan included using the cleaned dataset of county-year hazard data for predictive modeling. However, after performing the Exploratory Data Analysis (EDA), it was determined that several continuous variables were right-skewed, including population estimates, housing unit estimates and property loss measures. Additionally, the correlation matrix showed overlapping between variables, indicating multicollinearity. Lastly, the originally planned construction of the target variable included variables that were also meant to be predictors, which caused concern for data leakage and needed to be reevaluated.

These findings caused the plan to shift, ensuring predictor variables remained separate from those used for target creation.The composite hazard score continues to be created from

tornado frequency, intensity and property loss variables while the predictors focus on population, housing units, median home value and tornado characteristics (width and length). This is designed to avoid data leakage.

Multi-collinearity was observed with some variables, such as population and housing. During modeling, these will be evaluated to reduce multicollinearity effects.

2.2 Preprocessing

Several preprocessing steps were taken to ensure the dataset was ready for predictive modeling. The dataset was filtered by year to only retain observations between 1979-2024. This aligns with the original research data from Gensini and Brooks. The tornado events were then aggregated to a county-year level so that each observation represented one county in one year. This allows alignment with predicting which counties will flag as high hazard the following year using the target variable.

Data cleaning also included handling missing or invalid variables. The variable mag, which represents the F/EF-Scale 0-5, also included observations for -9 or unknown. These were replaced as missing (NA) so they do not distort averages or max values of other engineered features. These NA values will contribute to inevitable missingness, however, the decision was nevessary to ensure accuracy with the dataset.

During preprocessing, it was determined that the property loss variable required an additional correction after a closer review of the NOAA data documentation. Prior to 1996, tornado property loss was recorded using categorization by dollar amount where codes represent damage ranges: 0 = unknown or blank, 1 < $50, 2 = $50-$400, 3 = $500-$5,000, etc. Anything from 1996 onward includes the tornado property damage in millions of dollars. As 0 represents

unknown rather than a true zero, they were also replaced as missing (NA) to avoid distorting averages or max values. These will also contribute to missingness. To ensure consistency, pre-1996 categorical values were converted into approximate midpoint values in millions of dollars and post-1996 values remained their original value. This allowed for a consistent reporting of property loss for aggregation and modeling across the full timeframe.

2.3 Feature Engineering

Feature engineering was performed to capture additional dimensions of tornado hazard and exposure. Additional hazard-related features were added including average and maximum tornado intensity, count of strong tornadoes (F/EF-3 and greater), property loss per tornado and average property loss. Features that were inadvertently omitted during EDA were included to capture tornado characteristics for average length and width.

As there are vast differences in insurance property exposure and population density amongst these regions, normalized variables were created to help account for these differences. Tornadoes per 100,000 population and tornadoes per 10,000 housing units will allow for more meaningful comparisons between counties.

Prior to merging the cleaned data with the American Community Survey data, regional definitions were created defining the states within Tornado Alley and Dixie Alley using state FIPS codes allowing for feature creation classifying the regions.

2.4 Target Variable

The binary target variable was created for prediction of high tornado hazard next year. The composite score was created by averaging standardized measures of tornado frequency, intensity, strong tornadoes and log-transformed property loss. All variables were then standardized using z-scores to ensure comparability across differing scales. Any composite scores at or above the 75th percentile will be flagged as 1 = high hazard next year. Anything under this threshold will be labeled as 0 = not high hazard next year. This was also shifted forward one year with a lead function allowing the target variable to look at future hazard conditions.

2.5 Variable Summary Tables

Review of continuous variables including mean, median, standard deviation, min, max and skewness was performed as well as categorical variables for county and percent. From the continuous variable table, it shows significant right-skewing for population, housing unit estimates and property loss. This shows that most counties have a small amount of property damage due to tornado events while others reported large losses, which will be important for accurate prediction of future hazard counties. Tornado frequency is low but extremely variable, meaning most county-years are experiencing around one tornado but some also show high activity.

Log transformations were applied to variables that were highly right-skewed, including population, housing and property loss and are included in the Continuous Variables summary table. Population skewness decreased from over 11 to 0.318 after transformation. This was extremely important to perform for logistic regression modeling. XG Boost and random forest do not require these transformations so both transformed and untransformed variables will be considered depending on the predictive modeling method.

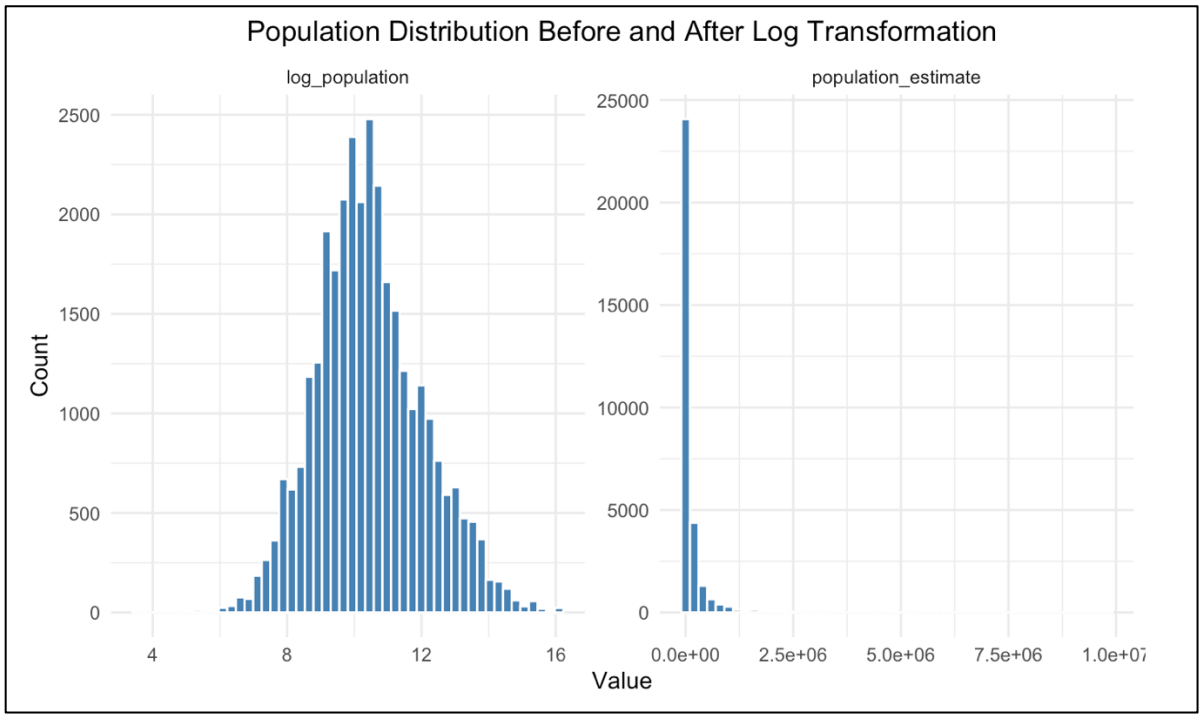

Figure 1: Histogram comparing population distribution before and after log transformation.

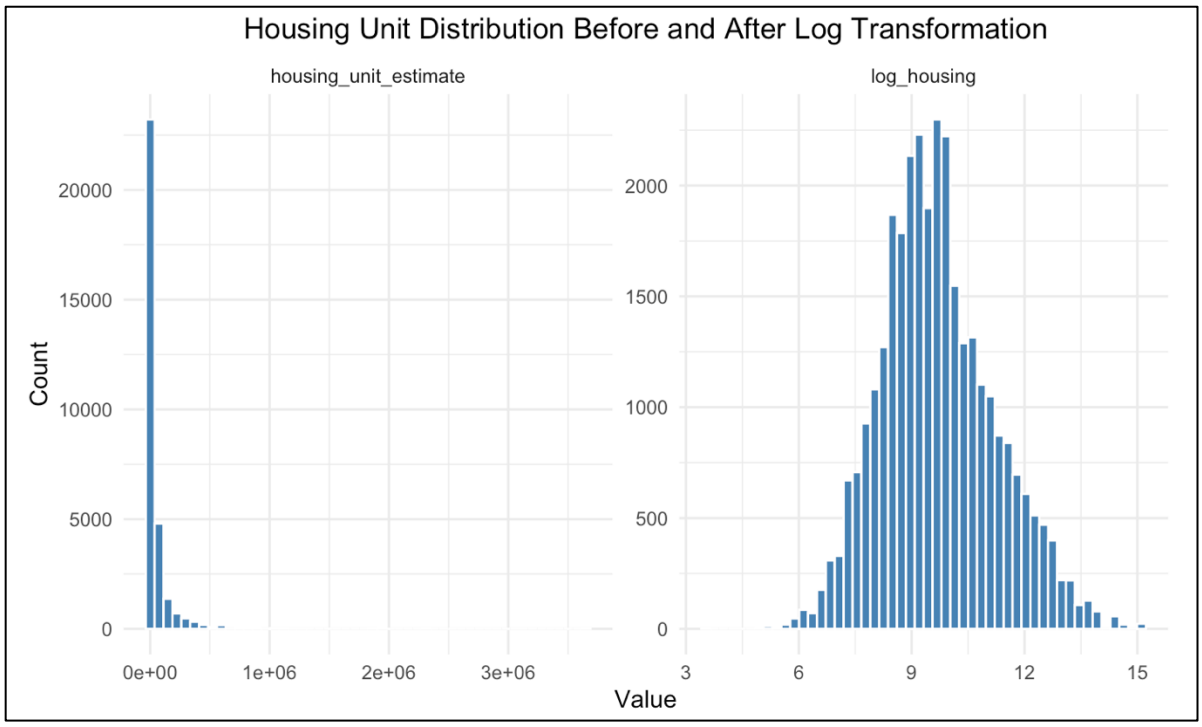

Figure 2: Histogram comparing housing unit distribution before and after log transformation.

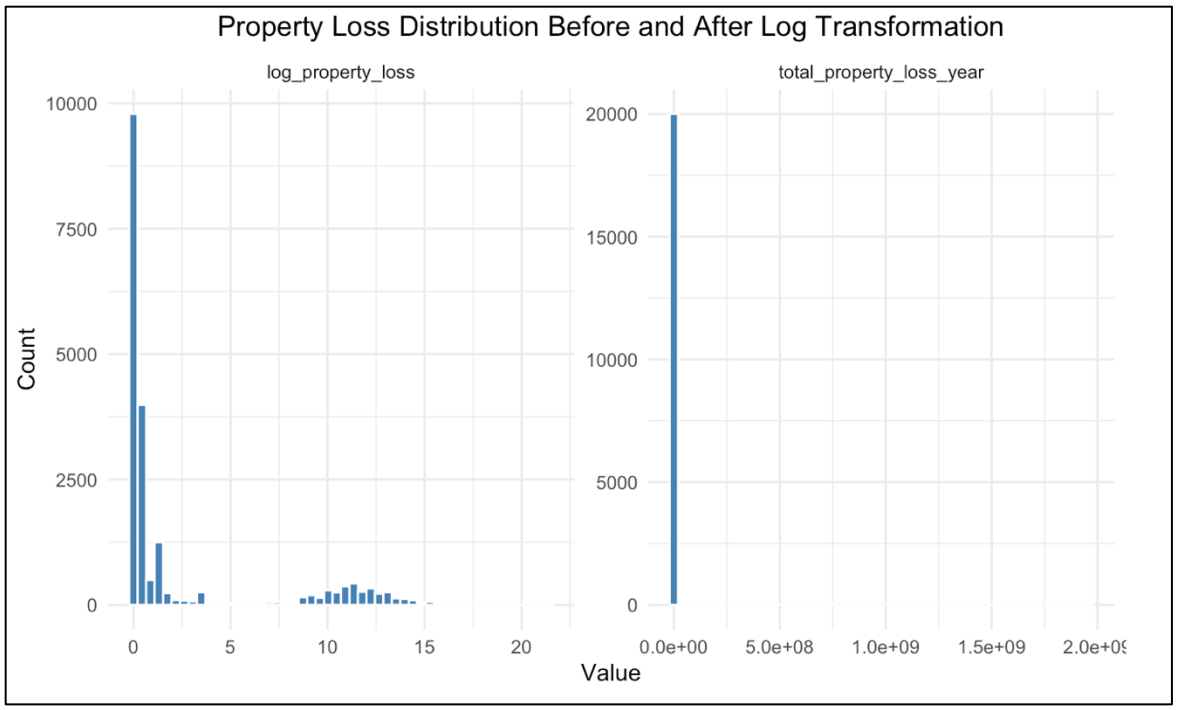

Figure 3: Histogram comparing property loss distribution before and after log transformation.

	Variable	Mean	Median	Standard_Deviation	Min	Max	Skewness
population_estimate	population_estimate	1.400886e+05	3.098600e+04	4.500747e+05	3.300000e+01	9.808667e+06	11.2694805
housing_unit_estimate	housing_unit_estimate	5.979875e+04	1.438600e+04	1.795066e+05	3.300000e+01	3.650466e+06	10.2176428
medvalue_estimate	medvalue_estimate	2.089561e+05	1.761000e+05	1.140000e+05	4.860000e+04	1.633900e+06	2.5468418
tornado_count_county_year	tornado_count_county_year	1.668815e+00	1.000000e+00	1.314475e+00	1.000000e+00	4.300000e+01	4.7953200
average_intensity_county	average_intensity_county	6.452601e-01	5.000000e-01	7.550947e-01	0.000000e+00	5.000000e+00	1.3030436
max_intensity_county	max_intensity_county	7.991814e-01	1.000000e+00	8.952859e-01	0.000000e+00	5.000000e+00	1.1291746
total_property_loss_year	total_property_loss_year	5.537277e+05	2.600000e-01	1.919971e+07	2.500000e-05	1.960040e+09	80.1712961
average_property_loss	average_property_loss	3.295584e+05	1.512500e-01	1.190645e+07	2.500000e-05	1.504000e+09	104.4075018
strong_tornado_count	strong_tornado_count	5.449000e-02	0.000000e+00	2.475277e-01	0.000000e+00	4.000000e+00	5.1733676
property_loss_per_tornado	property_loss_per_tornado	2.967931e+05	1.375000e-01	1.171642e+07	5.000000e-06	1.504000e+09	109.1299497
tornadoes_per_100k_population	tornadoes_per_100k_population	1.442167e+01	4.459508e+00	5.679958e+01	1.019507e-02	6.060606e+03	51.8640676
tornadoes_per_10k_housing_units	tornadoes_per_10k_housing_units	2.760752e+00	9.537434e-01	7.817134e+00	2.739376e-03	6.060606e+02	24.8121286
avg_length	avg_length	3.175988e+00	1.075000e+00	5.696500e+00	0.000000e+00	1.240000e+02	5.1672830
avg_width	avg_width	1.165054e+02	5.000000e+01	1.869472e+02	0.000000e+00	3.330000e+03	4.5387559
composite_hazard_score	composite_hazard_score	-2.285322e-02	-1.121555e-01	6.618299e-01	-6.190355e-01	8.129702e+00	1.9791569
log_property_loss	log_property_loss	2.411107e+00	2.311117e-01	4.446913e+00	2.499969e-05	2.139623e+01	1.7292479
log_population	log_population	1.045723e+01	1.034132e+01	1.579053e+00	3.526361e+00	1.609878e+01	0.3185034
log_housing	log_housing	9.694721e+00	9.574080e+00	1.517297e+00	3.526361e+00	1.511037e+01	0.3671376

Figure 4: Chart of Continuous Variables summary table

	Variable	Category	Count	Percent
1	high_tornado_hazard	0	23853	75.00
2	high_tornado_hazard	1	7951	25.00
3	high_tornado_hazard_next_year	0	21586	67.87
4	high_tornado_hazard_next_year	1	7208	22.66
5	high_tornado_hazard_next_year	NA	3010	9.46
6	region_category	D	6395	20.11
7	region_category	O	17566	55.23
8	region_category	T	7843	24.66

Figure 5: Chart of Categorical Variables summary table

2.6 Modeling Plan

The supervised methods for feature engineering will be used to select the most relevant features for the model and refined based on model performance by adding, removing or modifying features for improvement. The predictive modeling began with logistic regression, which will serve as the baseline model, as it supports the binary classification of the target variable. Feature importance for the this model will look to coefficient magnitude with L1 Regularization (Lasso). Additionally, random forest and XG Boost were utilized as flexible models to analyze if performance improves when compared to logistic regression. Random forest was selected due to the binary classification and less susceptibility to overfitting and better generalization. Feature importance will be evaluated on metrics with gini importance. XG Boost was selected due to parameters to prevent overfitting and higher predictive accuracy (Geeks for Geeks, 2025). Feature importance was utilized with Gain Importance.

The data was split using a time-based approach for county-year observations. The training set will include the earlier years of 1979-2014 and the testing set will use the later years of 2015-2024. This will help ensure the model is evaluated on future data.

After the models are built, they will use the test set to see how well it performs using recall and Precision-Recall Area Under the Curve, PR-AUC. Recall was prioritized as the primary evaluation metric to ensure identification of high-risk counties. This is essential for the stakeholder as failure to accurately identify high-risk counties could contribute to the continuation of underwriting in high insurance exposure areas. Confusion matrices will be used to provide a visualization of model performance.

The training set will also consider subgroup-aware cross-validation as this will ensure that observations for Tornado Alley and Dixie Alley are included as well as high risk and not high risk counties to reduce bias. This will be performed with a stratified k-fold rather than a conventional k-fold. Hyperparameter tuning will be conducted using cross-validation (randomized search) to optimize the model performance and reduce the risk of overfitting. These will include adjusting the number of trees in random forest, depth in xgboost and lambda in Lasso.

As the data contains strong geographic structure, the model may overfit by learning spatial dependence across counties rather than relying on the intended variables and exposure variables. This will not be fully addressed in this capstone, however, future work could include incorporate spatial lagging of predictors to better account for geographic relationships.

Unsupervised methods, such as PCA, LDA and clustering, were considered but ultimately excluded from this project. The primary goal for this project is to identify high hazard counties

next year using a binary outcome. Clustering could be an exploratory approach, however, it is not necessary as the composite hazard score has been created for tornado risk.

3 Discussion and Next Steps

During the preprocessing stage, variables that were highly skewed were log-transformed for better logistic regression modeling and to help normalize the extreme values. During the target creation, it was carefully curated using variables that would not also be used as predictors to ensure data leakage was avoided. Feature engineering included normalizing variables for tornados per 100,000 population and tornadoes per 10,000 housing units. This allows for better comparison across counties with varying levels of exposure. This will assist with identification of regions that are experiencing higher tornado hazard.

The next step of this project focuses on predictive modeling. Logistic regression will be used as the baseline model then will be compared to random forest and XG Boost. The model evaluation will prioritize recall and PR-AUC on the test set. Any feature adjustments will be performed depending on model performance. This is essential for stakeholders to be able to accurately identify high-risk counties allowing for informed underwriting decisions.

4 Github

https://github.com/jill-baker/tornado-insurance-risk-repo

5 Appendix

The Data Dictionary below includes identification of key raw, engineered and target variables in the final county-year tornado hazard and insurance exposure dataset. The target variable, high_tornado_hazard_next_year, was created from the hazard-related variables in the

dataset: tornado_count_county_year, average_intensity_county, max_intensity_county, total_property_loss_year and strong_tornado_count.

Column Name	Data Type	Description
om	Integer	Tornado number. Count of tornadoes during given year from NOAA dataset
yr	Integer	Year of tornado event (1950-2024)
mo	Integer	Month of tornado event, (1-12)
dy	Integer	Day of tornado event, (1-31)
date	Date	Date of tornado event in yyy-mm-dd format
st	String	State (two-letter postal abbreviation) of tornado event
mag	Integer	F-Scale (EF-Scale after 01/07) Values 0, 1, 2, 3, 4, 5, -9 (-9=unknown)
loss	Numeric	Estimated property loss in millions of dollars. 0 or blank = unknown, not $0
len	Numeric	Length of tornado in miles
wid	Integer	Width of tornado in yards
county_fips	String	Federal Information Processing Standards code for county, full 5 digit code (first 2 digits for state, last 3 digits for county).
county_name_state	String	Full county name and state
population_estimate	Integer	**Predictor.** Present-day exposure variable. Estimated total population in county
housing_unit_estimate	Integer	**Predictor.** Present-day exposure variable. Estimated total number of housing units in the county
medvalue_estimate	Numeric	**Predictor.** Present-day exposure variable. Estimated median home value in the county (in dollars)
region_category	String	**Predictor.** Indicates if county is located in Dixie Alley, Tornado Alley or Other. Dixie Alley = D, Tornado Alley = T, Other = O.
high_tornado_hazard_next_year	Binary	**Target variable.** Indicates whether a county is classified as high tornado hazard in the following year based on county's forward-shifted composite hazard classification (1=high tornado hazard, 0=not high tornado hazard)
tornado_count_county_year	Integer	*Target creation hazard variable.* Total number of tornadoes per county for a given year

average_intensity_county	Numeric	*Target creation hazard variable.* Average tornado intensity (F/EF-Scale) for tornadoes in county for a given year.
max_intensity_county	Integer	*Target creation hazard variable.* The maximum/highest observed tornado intensity per county in given year
total_property_loss_year	Numeric	*Target creation hazard variable.* The total property loss with all occurred tornadoes per county in given year
average_property_loss	Numeric	Hazard variable. The average property loss for tornadoes in county for a given year
strong_tornado_count	Integer	*Target creation hazard variable.* Count of tornadoes in county-year where the tornado intensity (mag) is >=3
property_loss_per_tornado	Numeric	Calculation of total_property_loss_year / tornado_count_county_year
tornadoes_per_100k_population	Numeric	Normalized measure: Calculation of tornado_count_county_year / population_estimate * 100000
tornadoes_per_10k_housing_units	Numeric	Normalized measure: Calculation of tornado_count_county_year / housing_unit_estimate * 10000
avg_length	Numeric	**Predictor.** Average length in miles for tornadoes per county
avg_width	Numeric	**Predictor.** Average width in yards for tornadoes per county
composite_hazard_score	Numeric	Composite index created by averaging standardized tornado frequency, intensity, and loss variables. Used to classify high hazard counties.
high_tornado_hazard	Binary	Variable used to create the predictive target high_tornado_hazard_next_year and represents the current-year classification
log_property_loss	Numeric	Logged transformation of total_property_loss_year
log_population	Numeric	Logged transformation of population_estimate
log_housing	Numeric	Logged transformation of housing_unit_estimate
z_avg_intensity	Numeric	Z-Score from average_intensity_county for target variable creation: high_tornado_hazard_next_year
z_max_intensity	Numeric	Z-Score from max_intensity_county for target variable creation: high_tornado_hazard_next_year

z_strong_count	Numeric	Z-Score from strong_tornado_count for target variable creation: high_tornado_hazard_next_year
z_property_loss	Numeric	Z-Score from log_property_loss for target variable creation: high_tornado_hazard_next_year
z_tornado_count	Numeric	Z-Score from tornado_count_county_year for target variable creation: high_tornado_hazard_next_year

6 References & Data Sets

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